

1 Gain law

(bars denote normalized quantities: lengths by R , $\bar{a} = a/a_o$, $\bar{f} = a_o f$; $0 < \bar{a} < 1$; $e_\theta = \theta - \hat{\theta}$)

The slope is read off the gain state alone. No parameter appears in the law.

$$\bar{a}'(\bar{a}) := \frac{d\bar{a}}{d\bar{w}} = (1 - \bar{a})^2$$

$$a_w = a_o \bar{a}_w, \quad a_o = \frac{\Delta t}{\gamma_N R} \quad (a_o \text{ fixed, not a state})$$

Integrated form (used for seeding only; $\bar{w}_0$ is the integration constant, i.e. setting $\hat{a}_w[0]$ IS setting the wall position)

$$\bar{a}(\bar{w}) = 1 - \frac{1}{\bar{w} - \bar{w}_0}, \quad \frac{\partial \bar{a}}{\partial \bar{w}_0} = -\bar{a}' \quad (\text{exact})$$

Declared truncation. The truth is not a member of this one-curve family. With $b_{\text{state}}(\bar{w}) := (1 - \bar{a}_{\text{true}})^2 / \bar{a}'_{\text{true}}$, measured offline on $\bar{w} - \bar{w}_s \in [0.1, 25]$:

$$\frac{\bar{a}'(\bar{a}_{\text{true}})}{\bar{a}'_{\text{true}}} - 1 = b_{\text{state}}(\bar{w}) - 1 \quad \begin{array}{l} 13.05 \% \text{ RMS, } 15.34 \% \text{ sup} \\ \text{range } [+7.79, +15.34] \%, \text{ **one-signed**} \\ \text{far field } \rightarrow +1/8 \text{ exactly } (b_{\text{state}} \rightarrow 9/8) \end{array}$$

This error is what derivation (b) asks the disturbance to absorb. Derivation (a) carries it with no container at all.

2 Gain rate and Jacobians

$$\frac{\partial \bar{a}'}{\partial \bar{a}} = -2(1 - \bar{a}) \quad (< 0 \text{ always})$$

$\bar{a}'$ depends on the state $\bar{a}_w$, so this partial is not zero. It propagates into S6 row 4 and into S7. Both derivations share it.

In (b) the disturbance enters the state equation additively, so its Jacobian is

$$\frac{\partial}{\partial \delta a} \left[\bar{a}_w[k] + (1 - \bar{a}_w[k])^2 \left(\Delta \bar{w}_d[k] + (1 - \lambda_c) \delta \bar{w}_3[k] + F_{dw}[k] e_{\bar{a}_w}[k] + \varepsilon_w[k] \right) + \delta a[k] \right] = 1 \quad (\text{constant; independent of } s)$$

3 True dynamics

$$(\delta \bar{w}_j[k] := \delta \bar{w}[k - d + j - 1], \quad j = 1, \dots, d + 1, \quad d = 2)$$

$$\bar{w}[k] = \bar{w}_d[k] - \delta \bar{w}_3[k]$$

$$\bar{w}_d[k + 1] = \bar{w}_d[k] + \Delta \bar{w}_d[k]$$

$$\delta \bar{w}_1[k + 1] = \delta \bar{w}_2[k]$$

$$\delta \bar{w}_2[k + 1] = \delta \bar{w}_3[k]$$

$$\delta \bar{w}_3[k + 1] = \lambda_c \delta \bar{w}_3[k] - F_{dw}[k] e_{\bar{a}_w}[k] - \varepsilon_w[k]$$

$$F_{dw}[k] := \bar{f}_{dw}[k] + (1 - \lambda_c) \sum_{i=1}^d \bar{f}_{dw}[k - i]$$

$$\varepsilon_w[k] := \bar{a}_w[k] \bar{f}_T[k] + (1 - \lambda_c) \sum_{i=1}^d \bar{a}_w[k - i] \bar{f}_T[k - i] + (1 - \lambda_c) n_w[k]$$

The gain follows the TRUE slope, which is not the law:

$$\bar{a}_w[k + 1] = \bar{a}_w[k] + \bar{a}'_{\text{true}}[k] \left[\Delta \bar{w}_d[k] + (1 - \lambda_c) \delta \bar{w}_3[k] + F_{dw}[k] e_{\bar{a}_w}[k] + \varepsilon_w[k] \right]$$

(a) baseline: no disturbance

The law is used as if it were the truth. The discrepancy has no container and is carried as the declared truncation of S1.

$$\bar{a}_w[k + 1] = \bar{a}_w[k] + (1 - \bar{a}_w[k])^2 \left[\Delta \bar{w}_d[k] + (1 - \lambda_c) \delta \bar{w}_3[k] + F_{dw}[k] e_{\bar{a}_w}[k] + \varepsilon_w[k] \right]$$

(b) with δa : the disturbance is DEFINED by the mismatch

δa is not postulated. It is whatever makes the law-based propagation exact:

$$\bar{a}_w[k + 1] = \bar{a}_w[k] + (1 - \bar{a}_w[k])^2 \left[\Delta \bar{w}_d[k] + (1 - \lambda_c) \delta \bar{w}_3[k] + F_{dw}[k] e_{\bar{a}_w}[k] + \varepsilon_w[k] \right] + \delta a[k]$$

Subtracting the true line above gives δa in closed form:

$$\begin{aligned} \delta a[k] &= \left[\bar{a}'_{\text{true}}[k] - (1 - \bar{a}_w[k])^2 \right] \left[\Delta \bar{w}_d[k] + (1 - \lambda_c) \delta \bar{w}_3[k] + F_{dw}[k] e_{\bar{a}_w}[k] + \varepsilon_w[k] \right] \\ &= (1 - \bar{a}_w[k])^2 \left[\frac{1}{b_{\text{state}}(\bar{w}[k])} - 1 \right] \left[\Delta \bar{w}_d[k] + (1 - \lambda_c) \delta \bar{w}_3[k] + F_{dw}[k] e_{\bar{a}_w}[k] + \varepsilon_w[k] \right] \end{aligned}$$

Three properties follow directly and are used in S8:

- (i) $\delta a[k]$ is proportional to the increment: it is ZERO during a hold.
- (ii) $1/b_{\text{state}} - 1 \in [-0.133, -0.072]$ is one-signed, but δa is NOT:
 $\text{sign}(\delta a) = -\text{sign}(\text{increment})$, so it reverses with the direction of travel.
 Measured on the canonical scenario: $\sum_k \delta a = +0.0680$ (descent), -0.0401 (ascent),
 and EXACTLY ZERO over a closed oscillation cycle — both $\bar{a}'_{\text{true}} d\bar{w}$ and
 $(1 - \bar{a}_{\text{true}})^2 d\bar{w}$ are exact differentials in $\bar{w}$.
**The whole disturbance is acquired on the descent; the oscillation offers
 a constant- δa model nothing to lock onto.**
- (iii) $(1 - \bar{a}_w)^2$ spans $0.83 \rightarrow 0.0019$: δa varies over three decades.

The process model of S4 nevertheless declares

$$\delta a[k+1] = \delta a[k]$$

This is a modelling CLAIM, not a consequence of the above. S8 prices what the claim omits.

4 Process model

(a) baseline

$$\mathbf{x}[k] = [\delta \bar{w}_1, \delta \bar{w}_2, \delta \bar{w}_3, \bar{a}_w]^\top [k] \quad (d=2)$$

$$\delta \bar{w}_1[k+1] = \delta \bar{w}_2[k]$$

$$\delta \bar{w}_2[k+1] = \delta \bar{w}_3[k]$$

$$\delta \bar{w}_3[k+1] = \lambda_c \delta \bar{w}_3[k] - F_{dw}[k] e_{\bar{a}_w}[k] - \varepsilon_w[k]$$

$$\bar{a}_w[k+1] = \bar{a}_w[k] + (1 - \bar{a}_w[k])^2 \left[\Delta \bar{w}_d[k] + (1 - \lambda_c) \delta \bar{w}_3[k] + F_{dw}[k] e_{\bar{a}_w}[k] + \varepsilon_w[k] \right]$$

(b) with δa

$$\mathbf{x}[k] = [\delta \bar{w}_1, \delta \bar{w}_2, \delta \bar{w}_3, \bar{a}_w, \delta a]^\top [k] \quad (d=2)$$

$$\delta \bar{w}_1[k+1] = \delta \bar{w}_2[k]$$

$$\delta \bar{w}_2[k+1] = \delta \bar{w}_3[k]$$

$$\delta \bar{w}_3[k+1] = \lambda_c \delta \bar{w}_3[k] - F_{dw}[k] e_{\bar{a}_w}[k] - \varepsilon_w[k]$$

$$\bar{a}_w[k+1] = \bar{a}_w[k] + (1 - \bar{a}_w[k])^2 \left[\Delta \bar{w}_d[k] + (1 - \lambda_c) \delta \bar{w}_3[k] + F_{dw}[k] e_{\bar{a}_w}[k] + \varepsilon_w[k] \right] + \delta a[k]$$

$$\delta a[k+1] = \delta a[k]$$

5 Estimator

(a) baseline

$$\begin{aligned}
\delta \hat{w}_1[k+1] &= \delta \hat{w}_2[k] \\
\delta \hat{w}_2[k+1] &= \delta \hat{w}_3[k] \\
\delta \hat{w}_3[k+1] &= \lambda_c \delta \hat{w}_3[k] \\
\hat{a}_w[k+1] &= \hat{a}_w[k] + (1 - \hat{a}_w[k])^2 \left[\Delta \bar{w}_d[k] + (1 - \lambda_c) \delta \hat{w}_3[k] \right]
\end{aligned}$$

(b) with δa

$$\begin{aligned}
\delta \hat{w}_1[k+1] &= \delta \hat{w}_2[k] \\
\delta \hat{w}_2[k+1] &= \delta \hat{w}_3[k] \\
\delta \hat{w}_3[k+1] &= \lambda_c \delta \hat{w}_3[k] \\
\hat{a}_w[k+1] &= \hat{a}_w[k] + (1 - \hat{a}_w[k])^2 \left[\Delta \bar{w}_d[k] + (1 - \lambda_c) \delta \hat{w}_3[k] \right] + \delta \hat{a}[k] \\
\delta \hat{a}[k+1] &= \delta \hat{a}[k]
\end{aligned}$$

The control law and the S7 measurement prediction both consume $\hat{a}_w[k]$ unchanged: δa enters the state equation, not the readout, so no separate “value used” quantity exists.

6 Error dynamics

(True – Estimator; coefficients at the estimate)

(a) baseline

$$\mathbf{e} = \left[e_{\delta \bar{w}_1}, e_{\delta \bar{w}_2}, e_{\delta \bar{w}_3}, e_{\bar{a}_w} \right]^\top$$

$$\begin{aligned}
e_{\delta \bar{w}_1}[k+1] &= e_{\delta \bar{w}_2}[k] \\
e_{\delta \bar{w}_2}[k+1] &= e_{\delta \bar{w}_3}[k] \\
e_{\delta \bar{w}_3}[k+1] &= \lambda_c e_{\delta \bar{w}_3}[k] - F_{dw}[k] e_{\bar{a}_w}[k] - \varepsilon_w[k] \\
e_{\bar{a}_w}[k+1] &= \left[1 + (1 - \hat{a}_w[k])^2 F_{dw}[k] - 2(1 - \hat{a}_w[k]) (\Delta \bar{w}_d[k] + (1 - \lambda_c) \delta \hat{w}_3[k]) \right] e_{\bar{a}_w}[k] \\
&\quad + (1 - \lambda_c) (1 - \hat{a}_w[k])^2 e_{\delta \bar{w}_3}[k] + (1 - \hat{a}_w[k])^2 \varepsilon_w[k]
\end{aligned}$$

$$F_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & \lambda_c & -F_{dw} \\ 0 & 0 & (1 - \lambda_c)(1 - \hat{a}_w)^2 & 1 + (1 - \hat{a}_w)^2 F_{dw} - 2(1 - \hat{a}_w)(\Delta \bar{w}_d + (1 - \lambda_c)\delta \hat{w}_3) \end{bmatrix}$$

(b) with δa

$$\mathbf{e} = [e_{\delta \bar{w}_1}, e_{\delta \bar{w}_2}, e_{\delta \bar{w}_3}, e_{\bar{a}_w}, e_{\delta a}]^\top$$

Row 4 gains one term and row 5 is an integrator of nothing:

$$\begin{aligned} e_{\bar{a}_w}[k+1] &= \left[1 + (1 - \hat{a}_w[k])^2 F_{dw}[k] - 2(1 - \hat{a}_w[k])(\Delta \bar{w}_d[k] + (1 - \lambda_c)\delta \hat{w}_3[k]) \right] e_{\bar{a}_w}[k] \\ &\quad + 1 \cdot e_{\delta a}[k] + (1 - \lambda_c)(1 - \hat{a}_w[k])^2 e_{\delta \bar{w}_3}[k] + (1 - \hat{a}_w[k])^2 \varepsilon_w[k] \\ e_{\delta a}[k+1] &= e_{\delta a}[k] \end{aligned}$$

$$F_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & \lambda_c & -F_{dw} & 0 \\ 0 & 0 & (1 - \lambda_c)(1 - \hat{a}_w)^2 & 1 + (1 - \hat{a}_w)^2 F_{dw} - 2(1 - \hat{a}_w)(\Delta \bar{w}_d + (1 - \lambda_c)\delta \hat{w}_3) & 1 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

The disturbance column is the constant 1: unlike a multiplicative parameter, its entry does not vanish at a turning point, does not vanish in a hold, and does not shrink in the far field. The control law is untouched, so $F_e(3, 5) = 0$: the loop coupling column is unchanged from (a).

Separability. A constant δa drives $e_{\bar{a}_w}$ as a RAMP, whereas an error in the gain state itself is a persistent OFFSET; the two are distinguishable by their time signature. This is the standard position-error / rate-bias pair.

$$\mathbf{q}[k] = [0, 0, -\varepsilon_w[k], (1 - \hat{a}_w[k])^2 \varepsilon_w[k], 0]^\top$$

7 Measurement and innovation

Measurement (d -step delay; $\bar{a}_{wm} = a_{wm}/a_o$). Identical in both derivations.

$$\begin{aligned} \nabla_d \bar{w}_d[k] &:= \bar{w}_d[k] - \bar{w}_d[k-d] \\ y_1[k] &= \delta \bar{w}_m[k] = \delta \bar{w}_1[k] + n_w[k] \\ y_2[k] &= \bar{a}_{wm}[k] = \bar{a}_w[k-d] + n_a[k] \end{aligned}$$

Delay back-off: the reading is the gain d steps ago, so walk it back along the trajectory by (slope) $\times$ (distance travelled),

$$\bar{a}_w[k-d] = \bar{a}_w[k] - (1 - \bar{a}_w[k])^2 \nabla_d \bar{w}_d[k] - d \delta a[k]$$

The third term exists only in (b) and is EXACT there: under $\delta a[k+1] = \delta a[k]$ the disturbance was applied d times over the delay, so walking the gain back must walk it back too.

$$(a) \quad y_2[k] = \bar{a}_w[k] - (1 - \bar{a}_w[k])^2 \nabla_d \bar{w}_d[k] + n_a[k]$$

$$(b) \quad y_2[k] = \bar{a}_w[k] - (1 - \bar{a}_w[k])^2 \nabla_d \bar{w}_d[k] - d \delta a[k] + n_a[k]$$

Linearization at the estimate. Because $\bar{a}'$ depends on the state, the partial is NOT 1:

$$\frac{\partial y_2}{\partial \bar{a}_w} = 1 + 2(1 - \hat{a}_w[k]) \nabla_d \bar{w}_d[k]$$

$$(a) \quad H[k] = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 + 2(1 - \hat{a}_w) \nabla_d \bar{w}_d \end{bmatrix}$$

$$(b) \quad H[k] = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 + 2(1 - \hat{a}_w) \nabla_d \bar{w}_d & -d \end{bmatrix} \quad \frac{\partial y_2}{\partial \delta a} = -d$$

The direct column is the CONSTANT $-d$: it does not collapse with position or with motion, unlike the multiplicative writing whose column carries a factor $(1 - \bar{a})^2 \nabla_d \bar{w}_d$. It is however small in magnitude against the retained term, so in practice δa is still seen mainly after it has accumulated into $\bar{a}_w$, i.e. through the cross-covariance $P(4, 5)$ that F_e builds.

Innovation

$$e_{y_1}[k] = y_1[k] - \delta \hat{w}_1[k] = e_{\delta \bar{w}_1}[k] + n_w[k]$$

$$e_{y_2}[k] = y_2[k] - \hat{a}_w[k] + (1 - \hat{a}_w[k])^2 \nabla_d \bar{w}_d[k] = \left[1 + 2(1 - \hat{a}_w[k]) \nabla_d \bar{w}_d[k] \right] e_{\bar{a}_w}[k] - d e_{\delta a}[k] + n_a[k] \quad (\text{the } -d e_{\delta a})$$

8 Process and measurement noise

Thermal block, identical in both derivations (only the current white thermal step enters Q):

$$q_3[k] = -\bar{a}_w[k] \bar{f}_T[k], \quad q_4[k] = (1 - \hat{a}_w[k])^2 \bar{a}_w[k] \bar{f}_T[k] = -(1 - \hat{a}_w[k])^2 q_3[k]$$

$$Q_{33}[k] = \frac{4k_B T a_o}{R} \bar{a}_w[k], \quad Q_{34}[k] = Q_{43}[k] = -(1 - \hat{\bar{a}}_w[k])^2 Q_{33}[k], \quad Q_{44}[k] = (1 - \hat{\bar{a}}_w[k])^4 Q_{33}[k]$$

Measurement noise $R = \text{diag}(R_1, R_2)$, unchanged from the sibling documents:

$$R_1 = \sigma_{n_w}^2, \quad R_2[k] = K_{\text{var}} I F_{\text{var}} (\bar{a}_w[k] + \xi)^2 + d Q_{44}[k], \quad K_{\text{var}} = \frac{2a_{\text{cov}}}{2 - a_{\text{cov}}}$$

The disturbance container Q_{55}

The process model of S4(b) claims $\delta a[k+1] = \delta a[k]$. S3(b) gives $\delta a[k]$ in closed form, so what the claim omits is computable, not assumed:

$$Q_{55}[k] = \text{Var}(\delta a[k+1] - \delta a[k])$$

with, writing $\Sigma[k] := \Delta \bar{w}_d[k] + (1 - \lambda_c) \delta \bar{w}_3[k] + F_{dw}[k] e_{\bar{a}_w}[k] + \varepsilon_w[k]$ for the increment of S3,

$$\delta a[k] = (1 - \bar{a}_w[k])^2 \left[\frac{1}{b_{\text{state}}(\bar{w}[k])} - 1 \right] \Sigma[k]$$

Because δa is proportional to Σ , the increment of δa is driven by the motion, not by time: Q_{55} is ZERO during a hold and grows with commanded speed. A time-invariant Q_{55} would therefore be the wrong container even if its magnitude were right.

FIRST ARM: $Q_{55} = 0$. This is the claim “no stochastic driving of δa ”; it is NOT the claim that δa is constant, which S3(b) already refutes. Its predicted consequence is that P_{55} decreases monotonically and $\delta \hat{a}$ freezes late in the run. That prediction is pre-registered here and is to be measured, not assumed.

PRE-REGISTERED CONSEQUENCE. S3(b)(i) says the true δa is zero in a hold, while $F_e(4, 5) = 1$ keeps the model injecting $\delta \hat{a}$ every step. With $Q_{55} = 0$ the covariance cannot reopen to disown it. So the final hold should show a spurious ramp in the gain error, of order the descent accumulation ($\sum_k \delta a = +0.068$ over 1.0 s) scaled by the hold duration (≈ 1.3 s), i.e. $\approx +0.09$ in $\bar{a}$ if unopposed by y_2 . Its presence, sign and magnitude are to be measured.

OPEN ITEM (not implemented in the first arm). δa inherits the frequency content of Σ , so on a sinusoidal trajectory it is dominated by the commanded frequency rather than being constant or white. The internal model principle then says the constant model $\delta a[k+1] = \delta a[k]$ cannot reject it asymptotically, and the derivable alternative is a two-state oscillator at the KNOWN commanded frequency. This is recorded so that the first arm’s residual can be read against it.

Seed and prior

$$\hat{\bar{a}}_w[0] = 1 - \frac{1}{\bar{w}[0] - \hat{\bar{w}}_0}, \quad \delta \hat{a}[0] = 0$$

$\hat{w}_0$ is the nominal wall; it enters here and nowhere else, since setting $\hat{a}_w[0]$ IS setting the integration constant.

$$\begin{aligned} \text{(a)} \quad P_{44}[0] &= (\hat{a}'[0])^2 P_{\bar{w}_0} + \text{floor}_a^2 \\ \text{(b)} \quad P_{44}[0] &= (\hat{a}'[0])^2 P_{\bar{w}_0} + \text{floor}_a^2, \quad P_{45}[0] = 0, \quad P_{55}[0] = P_{\delta a} \end{aligned}$$

$P_{45}[0] = 0$ because at $k = 0$ nothing has yet accumulated: the seed error and the disturbance are independent by construction. $P_{\delta a}$ is the only undetermined number; its scale follows from the level offset the disturbance is expected to remove divided by the number of steps over which it must remove it, and its contract is invariance (sweep it by $10\times$; the converged behaviour must not move).